

Matgeo Presentation - Problem 9.6.3

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Problem Statement

Find the area of the circle $x^2 + y^2 = 16$ exterior to the parabola $y^2 = 6x$.

Solution: Conic Parameters

The total area of the circle $x^2 + y^2 = 16$ is $A_{circle} = \pi r^2 = 16\pi$. First, we find the points of intersection. The conic parameters are:

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$$\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad f_1 = -16. \quad (1)$$

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For the parabola $y^2 - 6x = 0$:

$$\mathbf{v}_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad f_2 = 0. \quad (2)$$

Solution: Intersection of Conics

The intersection of the two conics is given by the equation:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (3)$$

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For a degenerate conic, the determinant of the quadratic part's matrix must be zero.

$$\det(\mathbf{V}_1 + \mu \mathbf{V}_2) = 0 \quad (4)$$

Solution: Finding Parameter μ

$$\det \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \mu \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{vmatrix} 1 & 0 \\ 0 & 1 + \mu \end{vmatrix} = 1(1 + \mu) = 0 \quad (5)$$

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Solution: Degenerate Conic Equation

Substituting $\mu = -1$ into equation (3):

$$\mathbf{x}^\top (\mathbf{V}_1 - \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (7)$$

$$\mathbf{x}^\top \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) \mathbf{x} + 2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} -3 \\ 0 \end{pmatrix} \right)^\top \mathbf{x} + (-16 - 0) = 0 \quad (8)$$

$$\mathbf{x}^\top \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} 3 \\ 0 \end{pmatrix}^\top \mathbf{x} - 16 = 0 \quad (9)$$

Solution: Solving for Intersection Points

Letting $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, the equation becomes:

$$(x \ y) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2(3 \ 0) \begin{pmatrix} x \\ y \end{pmatrix} - 16 = 0 \quad (10)$$

$$\implies x^2 + 6x - 16 = 0 \quad (11)$$

Solving for x : $(x + 8)(x - 2) = 0$, which gives $x = 2$ or $x = -8$.

Solution: Intersection Points

Substituting these values into $y^2 = 6x$:

$$\text{If } x = 2, y^2 = 12 \implies y = \pm\sqrt{12} = \pm 2\sqrt{3}.$$

If $x = -8$, $y^2 = -48$, which has no real solution for y .

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If $x = -8$, $y^2 = -48$, which has no real solution for y .

Hence, the points of intersection are:

$$\mathbf{a}_1 = \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ -2\sqrt{3} \end{pmatrix} \quad (12)$$

Solution: Area Calculation Setup

The area of the circle interior to the parabola is found by integration:

$$A_{interior} = 2 \left[\int_0^2 \sqrt{6x} \, dx + \int_2^4 \sqrt{16 - x^2} \, dx \right] \quad (13)$$

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First integral:

$$\begin{aligned} \int_0^2 \sqrt{6x} \, dx &= \sqrt{6} \left[\frac{2}{3} x^{3/2} \right]_0^2 = \frac{2\sqrt{6}}{3} (2\sqrt{2}) \\ &= \frac{4\sqrt{12}}{3} = \frac{8\sqrt{3}}{3} \end{aligned} \quad (14)$$

Solution: Area Calculation

Second integral (using trig substitution):

$$\begin{aligned}\int_2^4 \sqrt{16 - x^2} \, dx &= \left[\frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1} \left(\frac{x}{4} \right) \right]_2^4 \\&= (0 + 8 \sin^{-1}(1)) - \left(\sqrt{12} + 8 \sin^{-1} \left(\frac{1}{2} \right) \right) \\&= 8 \left(\frac{\pi}{2} \right) - \left(2\sqrt{3} + 8 \left(\frac{\pi}{6} \right) \right) = 4\pi - 2\sqrt{3} - \frac{4\pi}{3} \\&= \frac{8\pi}{3} - 2\sqrt{3}\end{aligned}\tag{15}$$

Solution: Final Area

Adding the results for $A_{interior}$:

$$\begin{aligned} A_{interior} &= 2 \left[\frac{8\sqrt{3}}{3} + \frac{8\pi}{3} - 2\sqrt{3} \right] \\ &= 2 \left[\frac{2\sqrt{3}}{3} + \frac{8\pi}{3} \right] = \frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \end{aligned} \tag{16}$$

Solution: Final Area

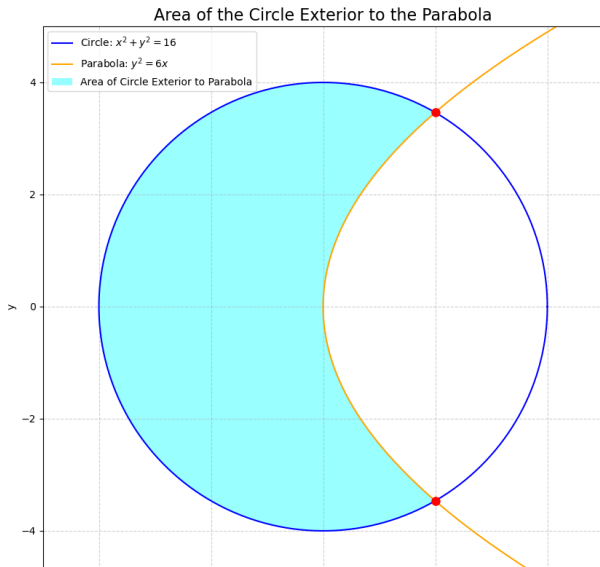
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The required area exterior to the parabola is $A_{exterior} = A_{circle} - A_{interior}$.

$$\begin{aligned} A_{exterior} &= 16\pi - \left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \right) \\ &= \frac{48\pi - 16\pi}{3} - \frac{4\sqrt{3}}{3} = \frac{32\pi - 4\sqrt{3}}{3} \end{aligned} \quad (17)$$

Plot



C Code: area_calculator.c

```
#include <math.h>

double integrand(double y) {
    if (16.0 - y * y < 0) {
        return 0.0;
    }
    return sqrt(16.0 - y * y) - (y * y / 6.0);
}

double calculate_common_area(double a, double b, int n) {
    double h = (b - a) / n;
    double sum = integrand(a) + integrand(b);

    for (int i = 1; i < n; i += 2) {
        sum += 4 * integrand(a + i * h);
    }

    for (int i = 2; i < n; i += 2) {
        sum += 2 * integrand(a + i * h);
    }
}
```

Python Code: main.py

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Polygon
import subprocess

try:
    subprocess.run(
        ['gcc', '-shared', '-fPIC', '-o', 'area_calculator.so', '
        area_calculator.c', '-lm'],
        check=True
    )
    lib = ctypes.CDLL('./area_calculator.so')
    lib.calculate_common_area.argtypes = [ctypes.c_double, ctypes
        .c_double, ctypes.c_int]
    lib.calculate_common_area.restype = ctypes.c_double
except (subprocess.CalledProcessError, OSError) as e:
    print(f"Error: {e}")
    exit()
```

Python Code: main.py (continued)

```
total_circle_area = 16 * np.pi
required_area = total_circle_area - common_area
analytical_area = (32 * np.pi - 4 * np.sqrt(3)) / 3

print(f"Total_Circle_Area: {total_circle_area:.8f}")
print(f"Common_Area (Numerical): {common_area:.8f}")
print(f"Required_Area (Numerical): {required_area:.8f}")
print(f"Required_Area (Analytical): {analytical_area:.8f}")

fig, ax = plt.subplots(figsize=(10, 10))
theta = np.linspace(0, 2 * np.pi, 200)
x_circle = 4 * np.cos(theta)
y_circle = 4 * np.sin(theta)
y_parabola_range = np.linspace(-6, 6, 200)
x_parabola = y_parabola_range**2 / 6

ax.plot(x_circle, y_circle, 'b-', label='Circle')
ax.plot(x_parabola, y_parabola_range, 'orange', label='Parabola')
ax.fill(x_circle, y_circle, 'cyan', alpha=0.4, label='Area_
    Exterior')
```

Python Code: verify.py

```
import numpy as np
from scipy.integrate import quad

def integrand(y):
    return np.sqrt(16 - y**2) - y**2 / 6

y_limit = 2 * np.sqrt(3)

common_area, error = quad(integrand, -y_limit, y_limit)

total_circle_area = 16 * np.pi
required_area = total_circle_area - common_area
analytical_area = (32 * np.pi - 4 * np.sqrt(3)) / 3

print("="*55)
print("VERIFICATION_USING_SCIPY")
print("="*55)
print(f"Common_Area_(SciPy): {common_area:.8f}")
print(f"Required_Area_(Calculated): {required_area:.8f}")
print(f"Required_Area_(Analytical): {analytical_area:.8f}")
```